The Pharma Innovation Journal 2026; 15(5): 09-12

www.ThePharmaJournal.com

The Pharma Innovation

ISSN (E): 2277-7695
ISSN (P): 2349-8242
Impact Factor (RJIF): 6.34
TPI 2026; 15(5): 09-12
© 2026 TPI
www.thepharmajournal.com
Received: 07-03-2026
Accepted: 09-04-2026

Devyani Ramola
Department of Genetics and Plant Breeding, College of Agriculture, G. B. Pant University of Agriculture & Technology, Pantnagar, Uttarakhand, India

Akriti Ballabh
Department of Animal Nutrition, College of Veterinary & Animal Sciences, G. B. Pant University of Agriculture & Technology, Pantnagar, Uttarakhand, India

Swati Semwal
Department of Genetics and Plant Breeding, College of Agriculture, G. B. Pant University of Agriculture & Technology, Pantnagar, Uttarakhand, India

Birendra Prasad
Department of Genetics and Plant Breeding, College of Agriculture, G. B. Pant University of Agriculture & Technology, Pantnagar, Uttarakhand, India

Corresponding Author:
Devyani Ramola
Department of Genetics and Plant Breeding, College of Agriculture, G. B. Pant University of Agriculture & Technology, Pantnagar, Uttarakhand, India

Evaluation of forage maize (*Zea mays* L.) genotypes for yield performance under north west zone conditions

Devyani Ramola, Akriti Ballabh, Swati Semwal and Birendra Prasad

DOI: https://www.doi.org/10.22271/tpi.2026.v15.i5a.26467

Abstract
Forage maize (*Zea mays* L.) is a critical component of the *Kharif* fodder production system in the North West Zone of India, valued for high biomass potential and adaptability to intensive cropping systems. The present study was conducted at, G.B. Pant University of Agriculture and Technology, Pantnagar, during *Kharif* 2024 under the All India Coordinated Research Project (AICRP) on Forage Crops and Utilization, to evaluate twelve forage maize genotypes comprising nine test entries and three national checks (African Tall, J-1006, and COHM-8) under the Second Advanced Varietal Trial (AVTM-2). The trial was laid out in a Randomised Block Design with three replications, and observations were recorded on green forage yield (GFY), dry matter yield (DMY), daily productivity indices, crude protein yield (CPY), crude protein percentage (CP%), plant height (PH), leaf-stem ratio (LSR), acid detergent fibre (ADF%), and neutral detergent fibre (NDF%). Significant genotypic variation was observed for yield traits, while fibre fractions showed limited variation. DFH 2022-1 recorded the highest GFY (704.5 q/ha) and DMY (167.7 q/ha), surpassing the best national check COHM-8 by 27.1% and 40.1% respectively, with the highest daily productivity indices (GFY/day: 10.67 q/ha/day; DMY/day: 2.54 q/ha/day), though its lowest CP% (4.29%) and highest ADF% (58.20%) indicate a yield-quality trade-off requiring consideration in feeding planning. KDFM-8 offered the most nutritively balanced profile, combining high GFY (672.3 q/ha), highest CPY (10.0 q/ha), moderate CP% (6.70%), and a favourable fibre profile (ADF: 56.10%; NDF: 60.9%). Both entries consistently and significantly outperformed all national checks for yield parameters, establishing their suitability for advancement in the AICRP pipeline under North West Zone conditions.

Keywords: Forage maize, green forage yield, dry matter yield, crude protein, ADF, NDF, yield-quality trade-off, North West Zone, AICRP

Introduction
Maize (*Zea mays* L.) is one of the most widely cultivated crops globally, valued simultaneously as a food, feed, and fodder crop. Among its uses, forage maize is particularly important for livestock feeding, owing to its high biomass productivity, rapid growth, and nutritive superiority over most other tropical annual fodder crops. Its C4 photosynthetic pathway enables efficient dry matter accumulation per unit area and time, making it a preferred choice for both green feeding and silage preparation for dairy livestock [1].

India faces a persistent deficit in quality green fodder estimated at over 35% of national requirement, directly constraining livestock productivity [2]. The North West Zone, with its intensive dairy farming systems, is particularly dependent on high-yielding *Kharif* fodder crops to meet this demand, and forage maize fits well into the short-duration, high-productivity cropping systems of this region. Genetic improvement through systematic varietal evaluation is therefore essential, as forage yield in maize is a complex trait governed by plant height, biomass accumulation, and crop duration, all subject to significant genotype × environment interaction [3]. Beyond yield, nutritive quality parameters crude protein (CP%), acid detergent fibre (ADF%), and neutral detergent fibre (NDF%) are equally critical, as ADF is inversely related to digestibility and NDF governs voluntary intake in ruminants [4]. A well-recognized challenge is the dilution effect, whereby rapid biomass accumulation in high-yielding genotypes reduces nitrogen concentration per unit dry matter, creating an inherent yield-quality trade-off [5].

The AICRP on Forage Crops and Utilization provides a structured multi-location framework for progressive evaluation of forage genotypes, advancing promising entries from Initial Varietal Trials through First and Second Advanced Varietal Trials (AVTM-2) before varietal recommendation. Entries at the AVTM-2 stage have already demonstrated yield superiority in earlier cycles and are evaluated for stability and adaptability across locations

Despite the availability of released checks such as African Tall, J-1006, and COHM-8, there remains a continuous need to identify genotypes with higher yield potential and better nutritive quality suited to specific zone conditions. In view of this, the present study was undertaken to evaluate advanced forage maize genotypes at Pantnagar for yield and quality traits, assess the yield-quality relationship across entries, and identify promising genotypes superior to national checks for advancement under North West Zone conditions.

Materials and methods

The present investigation was carried out at the N E Borlaug Crop Research Centre, G.B. Pant University of Agriculture and Technology, Pantnagar, during *Kharif* 2024 under the All India Coordinated Research Project (AICRP) on Forage Crops and Utilization, and analysis done at Forage Unit, CSKHPKV, Palampur. The Second Advanced Varietal Trial in Forage Maize (AVTM-2) comprised nine test entries - KDFM-8 (SKUAST, Srinagar), MAH 15-84 (UAS ZARS, Mandya), DFH 2022-1 (GBPUAT, Pantnagar), ADFM-4 (IARI, RRC Dharwad), PMC-14 (MPUAT, Udaipur), HPFM-12 (CSKHPKV, Palampur), PMC-16 (MPUAT, Udaipur), AFM-23 (AAU, Anand), ADFM-5 (IARI, RRC Dharwad) and three national checks - COHM-8, African Tall, J-1006.

The trial was laid out in a Randomised Block Design (RBD) with three replications. Each plot measured 4 m × 3 m, accommodating ten rows of 4 m length with a row-to-row spacing of 30 cm. Seed rate was maintained at 50 kg/ha (60 g/plot), with 4.5 kg seed per entry and per national check. A fertilizer dose of 80:40 kg N:P/ha was applied, of which 40:40 kg N:P/ha was given as basal at sowing and the remaining 40 kg N/ha was top-dressed 30 days after sowing.

Observations were recorded on green forage yield (GFY, q/ha), dry matter yield (DMY, q/ha), green forage yield per day (GFY/day, q/ha/day), dry matter yield per day (DMY/day, q/ha/day), crude protein yield (CPY, q/ha), crude protein percentage (CP%), plant height (PH, cm), leaf-stem ratio (LSR), acid detergent fibre (ADF%), and neutral detergent fibre (NDF%). Forage was harvested at the dough stage of grain development, Days to harvest from sowing to dough stage varied across entries, ranging from 66 days (DFH 2022-1, MAH 15-84, ADFM-4, COHM-8) to 71 days (African Tall), and were recorded individually for each entry for computation of daily productivity indices. which is considered optimum for maximizing both yield and nutritive value in forage maize (Filya, 2004). Quality parameters including CP%, ADF%, and NDF% were determined following standard AOAC procedures. Data were subjected to analysis of variance (ANOVA) and treatment means were compared using the critical difference (CD) at the 5% level of significance.

Results and discussion

Green Forage Yield and Dry Matter Yield

Significant differences were observed among the entries for green forage yield and dry matter yield, while the remaining traits (GFY/day, DMY/day, CPY, CP%, ADF%, NDF%, PH, and LSR) are reported as entry means across three replications (Table 1). Green forage yield (GFY) ranged from 380.1 q/ha (PMC-16) to 704.5 q/ha (DFH 2022-1), with a trial mean of 496.5 q/ha and CD at 5% of 41.4 q/ha (CV: 8.4%). DFH 2022-1 recorded the highest GFY, exceeding all three national checks - COHM-8 (554.3 q/ha), African Tall (404.5 q/ha), and J-1006 (403.6 q/ha) by 27.1%, 74.1%, and 74.6%, respectively. KDFM-8 (672.3 q/ha) was the second highest-yielding test entry, also significantly surpassing all national checks. Entries ADFM-4 (503.6 q/ha) and PMC-14 (484.6 q/ha) performed at par with the trial mean. The superior GFY of DFH 2022-1 and KDFM-8 can be partly attributed to their significantly taller plant stature (309.6 cm and 257.3 cm, respectively), which directly contributes to greater canopy interception and biomass accumulation [6]. The positive association between plant height and forage yield in maize has been consistently reported in earlier studies [7].

Dry matter yield (DMY), which is a more reliable index of actual forage worth than green yield alone [8], followed a similar trend. DFH 2022-1 again topped with 167.7 q/ha, followed by KDFM-8 (149.2 q/ha), both significantly higher than COHM-8 (119.7 q/ha), African Tall (93.4 q/ha), and J-1006 (88.9 q/ha). The trial mean DMY was 112.4 q/ha (CD at 5%: 12.8). Daily productivity indices, GFY/day (10.67 q/ha/day) and DMY/day (2.54 q/ha/day), were also highest for DFH 2022-1, reinforcing its superior biomass accumulation rate across the growing period. KDFM-8 recorded GFY/day of 10.03 and DMY/day of 2.23, the second highest values among all entries. High daily productivity is particularly important under the intensive cropping systems of the North West Zone, where turnaround time between crops is critical [9].

Crude Protein Yield and Crude Protein Percentage

Crude protein yield (CPY) ranged from 5.5 q/ha (African Tall) to 10.0 q/ha (KDFM-8), with a trial mean of 6.95 q/ha. KDFM-8 recorded the highest CPY despite DFH 2022-1 producing the greatest total biomass, a distinction that becomes meaningful when the CP% values are examined alongside yield. CP% varied considerably across entries, from 4.29% (DFH 2022-1) to 8.10% (J-1006), with a trial mean of 6.31%. This inverse pattern between GFY and CP%, where the highest-yielding genotype recorded the lowest CP% and the lower-yielding check recorded the highest CP% reflects the well-documented dilution effect in forage crops, whereby rapid biomass accumulation leads to a proportionate reduction in nitrogen concentration per unit dry matter. Such a trade-off between yield and protein concentration has been widely reported in forage maize and other annual fodder crops [10].

Importantly, however, KDFM-8 presents a favourable combination, high GFY (672.3 q/ha), the highest CPY (10.0 q/ha), and a reasonable CP% (6.70%), suggesting that it accumulates sufficient total protein alongside high biomass, making it nutritionally competitive despite the dilution effect. In practical terms, CPY rather than CP% alone is the more relevant measure for livestock feeding systems where total nutrient supply per unit area matters. DFH 2022-1, while lowest in CP% at 4.29%, still produced 7.2 q/ha CPY on account of its exceptionally high DMY, which remains within the range adequate for maintenance and moderate production in ruminants when supplemented appropriately [8].

Fibre Fractions: ADF and NDF

ADF% ranged from 55.40% (ADFM-4) to 58.20% (DFH 2022-1), with a trial mean of 56.41%, while NDF% ranged from 59.2% (AFM-23) to 65.8% (African Tall), with a mean of 62.38%. The range of variation in both ADF and NDF across entries was narrow, indicating limited genotypic diversity for fibre composition among these advanced nursery materials, which is not unexpected given that entries in AVTM-2 have already passed through prior screening cycles. ADF is

inversely related to dry matter digestibility, and NDF is negatively associated with voluntary intake in ruminants [11]. Values observed here (ADF ~55-58%; NDF ~59-66%) are typical for tropical forage maize evaluated at the dough stage and are broadly acceptable for ruminant feeding, though they indicate moderate digestibility [12]. It is noteworthy that DFH 2022-1, despite recording the highest biomass yield, also had the highest ADF% (58.20%) in the trial, and a relatively high NDF% (65.2%), which is a point of consideration from a digestibility standpoint. KDFM-8 in contrast showed a more favourable fibre profile (ADF: 56.10%; NDF: 60.9%) alongside its high yield, suggesting better cell wall quality relative to DFH 2022-1. The absence of any dramatic elevation in fibre content across test entries compared to national checks indicates that yield gains in advanced genotypes have not come at a major cost to fibre composition [10].

Leaf-Stem Ratio and Plant Height

LSR ranged from 0.35 (HPFM-12) to 0.66 (PMC-14), with a trial mean of 0.51. A higher LSR is generally considered desirable in forage crops as leaves are more digestible, higher in protein, and more palatable than stems [13]. PMC-14 (LSR: 0.66) and ADFM-5 (LSR: 0.59) recorded the highest LSR among test entries, while DFH 2022-1 (LSR: 0.57) and KDFM-8 (LSR: 0.42) showed contrasting values despite both being high-yielding. The relatively lower LSR of KDFM-8 in the context of its high biomass is understandable, as taller, higher-biomass genotypes tend to have proportionally larger stem fractions [6]. Plant height ranged from 211.9 cm (African Tall) to 309.6 cm (DFH 2022-1), with a mean of 250.4 cm. DFH 2022-1 was the tallest entry in the trial, and its height advantage (309.6 cm) over all checks was consistent with its dominant yield performance.

Overall Assessment

Taken together, the data reveal a clear differentiation among entries when yield and quality are considered simultaneously. DFH 2022-1 stands out as the highest-yielding genotype, significantly surpassing all national checks for GFY, DMY, and daily productivity; however, its relatively higher ADF% (58.20%) and lower CP% (4.29%) indicate a moderate quality profile that would need to be accounted for in feeding system planning. KDFM-8 offers a more balanced combination across traits, high GFY (672.3 q/ha), highest CPY (10.0 q/ha), moderate CP% (6.70%), and a more favourable fibre profile (ADF: 56.10%; NDF: 60.9%), making it the most nutritively well-rounded entry in the trial. Both entries significantly and consistently surpassed all national checks for yield parameters, underscoring their potential for advancement and zone-specific recommendation under North West Zone conditions. The results are consistent with findings from similar AICRP forage maize evaluations where newer hybrid genotypes have demonstrated marked superiority over open-pollinated checks such as African Tall and J-1006 [9].

Table 1: Performance of forage maize genotypes for yield and quality traits under AVTM-2, Kharif 2024, Pantnagar

Entries	GFY (q/ha)	DMY (q/ha)	GFY (q/ha/ day)	DMY (q/ha/ day)	CPY (q/ha)	CP %	PH (cm)	LSR	ADF %	NDF %
KDFM-8	672.3	149.2	10.03	2.23	10	6.7	257.3	0.42	56.1	60.9
MAH 15-84	494.4	106.8	7.49	1.62	7.2	6.74	246.5	0.46	55.55	64.3
DFH 2022-1	704.5	167.7	10.67	2.54	7.2	4.29	309.6	0.57	58.2	65.2
ADFM-4	503.6	110.3	7.63	1.67	7.3	6.62	303.2	0.49	55.4	62.9
PMC-14	484.6	112.4	6.92	1.61	6.6	5.87	241.7	0.66	56.65	60
HPFM-12	465.8	102.5	6.95	1.53	5.9	5.76	225.5	0.35	56.25	62
PMC-16	380.1	91.2	5.59	1.34	6.5	7.13	238.7	0.6	56.73	63
AFM-23	415.8	95.4	6.21	1.42	7.3	7.65	287.3	0.4	56.43	59.2
ADFM-5	474.5	111	6.78	1.59	6.2	5.59	240.6	0.59	56.62	65.2
COHM-8 (NC)	554.3	119.7	8.4	1.81	6.5	5.43	221.2	0.49	56.36	60.3
African Tall (NC)	404.5	93.4	5.7	1.32	5.5	5.89	211.9	0.53	57.02	65.8
J-1006 (NC)	403.6	88.9	6.02	1.33	7.2	8.1	220.9	0.54	55.58	59.3
Mean	496.5	112.4	7.37	1.67	6.95	6.31	250.4	0.51	56.41	62.38
CD at 5%	41.4	12.8	0.6	0.15	0.4	0.19	23.5	0.05	1.2	1.3
CV (%)	8.4	11.7	10.3	13.5	7.5	3.2	9.6	12.8	2.4	2.2

Abbreviations

GFY, Green Forage Yield; DMY, Dry Matter Yield; GFY/day, Green Forage Yield per day; DMY/day, Dry Matter Yield per day; CPY, Crude Protein Yield; CP%, Crude Protein Percentage; PH, Plant Height; LSR, Leaf-Stem Ratio; ADF%, Acid Detergent Fibre; NDF%, Neutral Detergent Fibre.

Conclusion

The present study evaluated nine forage maize genotypes against three national checks under the Second Advanced Varietal Trial (AVTM-2) of AICRP on Forage Crops and Utilization at Pantnagar during *Kharif* 2024. Significant genotypic variation was recorded for forage yield traits, while fibre fractions remained relatively stable across entries. DFH 2022-1 emerged as the highest-yielding genotype with GFY of 704.5 q/ha and DMY of 167.7 q/ha, surpassing all national checks by a wide margin, though its higher ADF% and lowest CP% (4.29%) indicate that its nutritive quality merits attention in feeding system design. KDFM-8, with the highest CPY (10.0 q/ha), a favourable fibre profile (ADF: 56.10%; NDF: 60.9%), and GFY of 672.3 q/ha, demonstrated the best overall balance between biomass productivity and nutritive quality among all entries. Both DFH 2022-1 and KDFM-8 consistently and significantly outperformed the national checks African Tall, J-1006, and COHM-8 for yield parameters, confirming the superiority of advanced hybrid genotypes over traditional open-pollinated varieties under the agro-climatic conditions of the North West Zone. These two entries are recommended for consideration for advancement in the AICRP pipeline and potential zone-specific recommendation, subject to confirmation of their performance across other locations within the zone.

References

1. Filya I. Nutritive value and aerobic stability of whole crop maize silage harvested at four stages of maturity. Anim Feed Sci Technol. 2004;116(1-2):141-150.
2. IGFRI. IGFRI vision 2050. ICAR-Indian Grassland and Fodder Research Institute, Jhansi; 2013.
3. Albrecht KA, Thomason WE. Corn silage management. In: Buxton DR, Muck RE, Harrison JH, editors. Silage science and technology. Madison: American Society of Agronomy; 2003. p. 521-558.
4. Van Soest PJ. Nutritional ecology of the ruminant. 2nd ed. Ithaca: Cornell University Press; 1994.
5. Ciampitti IA, Vyn TJ. Physiological perspectives of changes over time in maize yield dependency on nitrogen uptake and associated N efficiencies: a review. Field Crops Res. 2012;133:48-67.
6. Seydoşoğlu S. The effects of different row spacings on yield and quality traits of some maize varieties cultivated as second crop. Appl Ecol Environ Res. 2019;17(3):5759-5770.
7. Gözübenli H, Kilinç S, Şener O, Konuskan O. Effects of single and twin row planting on yield and yield components in maize. Asian J Plant Sci. 2004;3(2):203-206.
8. McDonald P, Edwards RA, Greenhalgh JFD, Morgan CA. Animal nutrition. 6th ed. London: Pearson Education; 2002.
9. Mahanta SK, Prasad S, Singh S. Forage production technologies for sustainable livestock production: an overview. Range Manag Agrofor. 2011;32(2):79-88.
10. Barrière Y, Guillet C, Goffner D, Pichon M. Genetic variation and breeding strategies for improved cell wall digestibility in annual forage crops: a review. Anim Res. 2003;52(3):193-228.
11. Mertens DR. Regulation of forage intake. In: Fahey GC Jr., editor. Forage quality, evaluation, and utilization. Madison: American Society of Agronomy; 1994. p. 450-493.
12. Djalovic I, Karnatam S, Srivastava RK. Silage maize as a potent candidate for sustainable animal husbandry development perspectives and strategies for genetic enhancement. Front Genet. 2023;14:1150132.
13. Jung HG, Allen MS. Characteristics of plant cell walls affecting intake and digestibility of forages by ruminants. J Anim Sci. 1995;73(9):2774-2790.